

Titanium electronic anode for water heaters

Anode with a mixed metal oxide (MMO) catalytic coating

The titanium electronic anode is a modern solution for protecting water heaters against internal corrosion. Unlike traditional sacrificial anodes, it has a core of high-strength titanium with a catalytic coating of mixed iridium and tantalum oxides (Ir-Ta): a longer-lasting and more economical solution.

The anode is powered by an electric current and continuously supplies the electrons that protect the steel tank against corrosion and extend the service life of the water heater. Because the anode is not consumed the way a sacrificial one is, it leaves no sediment, needs almost no servicing and works for more than 10 years. It also softens the water, reduces scale build-up and removes the unpleasant «rotten egg» sulphur smell from the tank.

1. Anode technical data

Anode model	Tank volume without a heat exchanger coil, l (gal.)		Anode thread to the customer's choice
	from	to	
Ti 200	50 (13)	200 (52)	3/4"-14 NPT, G 3/4", R 3/4" and other types
Ti 400	200 (52)	500 (132)	
Ti 800	500 (132)	1000 (264)	

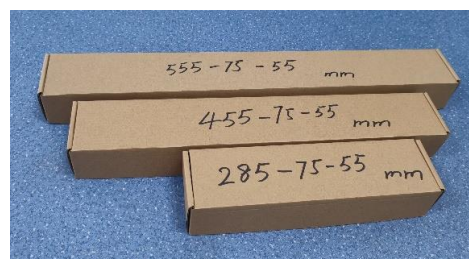
Manufacture to the customer's requirements is possible.



The set: anode and power supply unit

2. Packaging

Model	Box size	Weight
Ti 200	285 × 75 × 55 mm	317 g
Ti 400	455 × 75 × 55 mm	344 g
Ti 800	555 × 75 × 55 mm	368 g



Packaging of the three sizes

3. Power supply unit (potential controller)

3.1. Purpose

The power supply unit, also called the potential controller, measures and regulates the potential level, keeping the system working at its optimum. It runs on a 110–240 V mains supply and comes in two versions: **PE** — for enamelled carbon steel tanks and **PS** — for stainless steel tanks. Simplified versions without potential control are also available.

3.2. Comparison with a power supply unit without potential control

Feature	Power supply unit with potential control	Standard power supply unit
Protection mode	Maintains tank protection continuously; adjusts the output current according to the corrosion conditions	Constant output current, no potential control
Supply voltage	Universal range 110–240 V	Requires a separate version for each voltage
Tank type	PE version — enamelled carbon steel tanks; PS — stainless steel tanks	No split by tank type
Appearance		

3.3. Data of the power supply unit with potential control

Input	~ 110–240 V, 50–60 Hz, max. 0.03 A
Output	max. 10 V DC, max. 0.2 A
Ambient temperature	0 ... +70 °C
Protection rating	IP20
Indication	Green LED — everything is working correctly; red LED — a fault has been detected, the controller has stopped

3.4. Data of the standard power supply unit

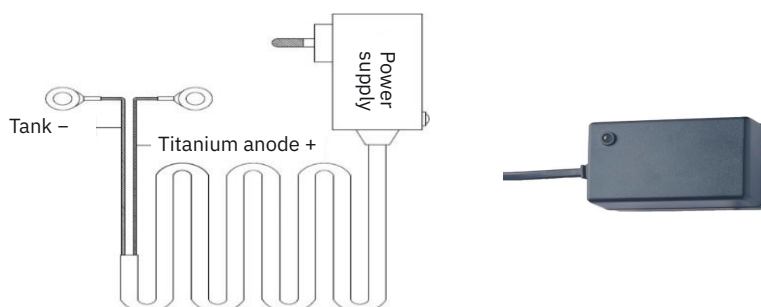
Input	~ 120 V / 240 V, 50/60 Hz
Output	24 V DC, 0.5 A

Manufacture to the customer's requirements is possible.

4. Installation and connection

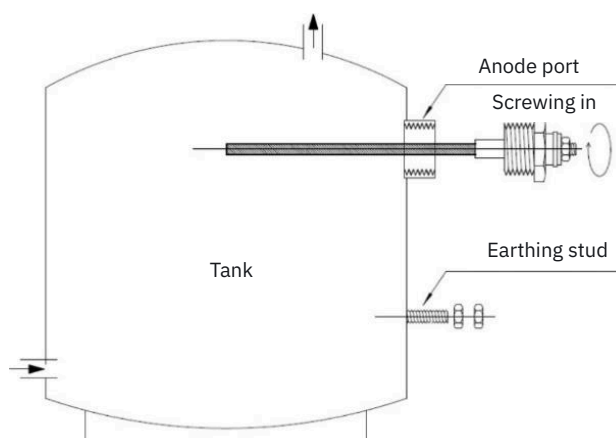
Step 1

Take the potential controller and the anode out of the packaging. Check the contents: anode, power supply unit with cable, two ring terminals.



Step 2

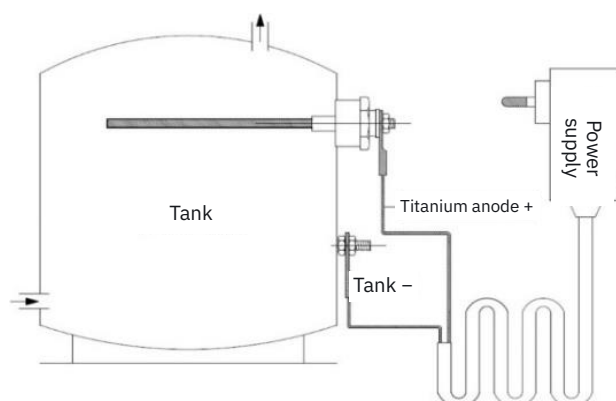
Fit the titanium anode into the tank: screw it into the threaded port and tighten it with a 27 mm spanner.



Step 3

Connect the wires: «+» to the titanium anode, «-» to the earthing stud of the tank through a ring terminal.

In the original manufacturer's manual this step is described, in error, by the same text as step 2.



Step 4

Plug the potential controller into the mains.

Step 5

Check that the controller is working correctly:

- the **green** LED is lit — everything is in order;
- the **red** LED is lit — a fault has been detected, the controller has stopped.

5. Warnings and remarks



WARNING 1

Wiring the leads the wrong way round will not protect the tank — on the contrary, it will **accelerate its corrosion!**



WARNING 2

Once fitted, the titanium anode **must not touch any metal** — the thermostat pocket, the tank wall, heat exchanger coils, the immersion heater and so on. Otherwise the anode will cause accelerated corrosion of those parts.



REMARK 1

Anodes with a G 3/4", R 3/4" or NPT 3/4"-14 thread are fitted with a **27 mm** spanner.



REMARK 2

On enamelled (glass-lined) carbon steel tanks the earthing stud may need to be cleaned of the scale formed during enamelling. This ensures a reliable electrical contact between the stud and the ring terminal of the «-» lead.